

Total no. of Pages 02

Roll no

3rd Sem.-B.Tech. .

END-SEMESTER EXAMINATION, NOV.-DEC.,2024

Course Code- EPMTTC04

Course Title- Mathematics for Machine Learning

Time- 03 Hours

Max. Marks-50

Note: - Attempt all five questions. Missing data/ information if any, may be suitably assumed & mentioned in the answer.

Q.No.	Attempt any 2 parts of the questions below	Marks	CO
1(a)	Let V be the set of all positive real no's with addition defined as $x(+)y = xy$ and scalar multiplication defined as $\alpha(.)x = x^\alpha$. Check whether V is a vector space.	05	CO1
1(b)	Let V and W be two vector spaces in R^3 . Let $T: V \rightarrow W$ be a linear transformation defined by $T(x, y, z) = (x + z, x + y, x + y + z)$. Find the matrix representation of T with respect to the basis $X = \left\{ \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\}$ in V and $Y = \left\{ \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \right\}$ in W .	05	CO1
1(c)	Find Kernel and Image of the linear transformation $T: R^4 \rightarrow R^3$ such that $T(x, y, z, w) = (x + y + w, z, y + 2w)$. Also verify Rank- Nullity Theorem/ Sylvester law.	05	CO1
2(a)	Determine whether $q(x, y, z) = x^2 + y^2 + 2xz + 4yz + 3z^2$ is positive definite. Also find the Eigen Values of the associated matrix.	05	CO2
2(b)	Consider the space of polynomials over R with inner product $\langle f, g \rangle = \int_{-1}^1 f(t)g(t)dt$. Find an orthogonal basis for the subspace $W = P_3(t)$ i.e. the space of all polynomials of degree less than or equal to 3, by applying the Gram- Schmidt Algorithm to $\{1, t, t^2, t^3\}$.	05	CO2
2(c)	Let $T: R^3 \rightarrow R^3$ be linear transformation defined by $T(x, y, z) = (x + 2y + 3z, x + 2y + 5z, 2x + 4y + 6z)$ and $V = \{(1, 1, 1), (2, 1, 2), (1, 2, 3)\}$. Show that the images of the vectors in V under T are linearly dependent.	05	CO2
3(a)	Find the singular value decompositions of the matrix $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -3 \end{bmatrix}$	05	CO3
3(b)	Find the polar decomposition of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ -2 & 1 & 0 \end{bmatrix}$	05	CO3
3(c)	Find Pseudo inverse of the matrix $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$	05	CO3
4(a)	Show that the vector field $\vec{V} = 3x^2y^2z^4\hat{i} + 2x^3yz^4\hat{j} + 4x^3y^2z^3\hat{k}$ is irrotational. Hence find a scalar function $\phi(x, y, z)$ such that $V = \text{grad}\phi$	05	CO4

4(b)	Use Divergence theorem to evaluate the surface integral $\iint_S \vec{V} \cdot \hat{n} dS$ where $\vec{V} = 2xy^2\hat{i} + x^2y\hat{j} + x^3\hat{k}$ and D is the region bounded by $z = \sqrt{x^2 + y^2}$ and the plane $z = 4$.	05	CO4
4(c)	Verify Green's Theorem by evaluating both integrals $\oint_C x^3 dy - y^3 dx$, C is the curve given by $x = 2\cos\theta$, $y = 2\sin\theta$ and $0 \leq \theta \leq 2\pi$	05	CO4
5(a)	Minimize: $f(x, y) = x^2 + y^2 + 60x$ Subject to: $x - 80 \geq 0$ $x + y - 120 \geq 0$	05	CO5
5(b)	Maximize: $x^m y^n z^p$ Subject to: $x + y + z - a = 0$	05	CO5
5(c)	Apply the method of Steepest Decent to the function $f(x, y) = x^2 - xy + y^2$, with the initial point $x_0 = (1, 0.5)$ such that $ f_{k+1} - f_k < 0.05$	05	CO5